

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree (S, FE) Examination May 2024 (2019 Scheme)

Course Code: CST415**Course Name: INTRODUCTION TO MOBILE COMPUTING****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| 1 | Explain different types of middleware and gateways required in the architecture of mobile computing. | (3) |
| 2 | Explain the major segments to support mobile computing function with diagram. | (3) |
| 3 | Compare and contrast Satellite systems-GEO, LEO and MEO. | (3) |
| 4 | Draw a call flow diagram for a theatre ticket booking system | (3) |
| 5 | Illustrate the billing and charging services in GPRS. | (3) |
| 6 | What are the various strengths of SMS? | (3) |
| 7 | What are the different types of Wireless LANs? | (3) |
| 8 | Explain about HIPERLAN | (3) |
| 9 | Explain Multimedia broadcast services | (3) |
| 10 | Explain the key features of TLS protocol. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- | | | |
|----|--|------|
| 11 | a) Describe the design considerations in mobile computing | (4) |
| | b) Explain the three-tier architecture of mobile computing with figure | (10) |

OR

- | | | |
|----|----------------------------|-----|
| 12 | a) Explain the terms | (8) |
| | (i) Client Context Manager | |
| | (ii) Policy Manager | |
| | (iii) Security Manager | |
| | (iv) Adaptability Manager | |

1000CST415122201

- b) Give examples of mobile computing applications (6)

Module II

- 13 a) Describe the different algorithms used for security and authentication in GSM. (7)
b) Why is multiple access important? With the help of suitable examples, explain the various multiple access techniques (7)

OR

- 14 a) Show how call routing is done in GSM. Give an example (7)
b) Explain the process of handover. How does handover differ from roaming? (7)

Module III

- 15 a) Explain GPRS architecture (10)
b) What are the various strengths of SMS? (4)

OR

- 16 a) Describe the SMS architecture. (10)
b) List out the applications of GPRS (4)

Module IV

- 17 a) Compare Wifi Vs 3G (8)
b) Explain about WLL Architecture (6)

OR

- 18 a) Explain Briefly WLAN architecture with figure (10)
b) What are the advantages of WLAN? (4)

Module V

- 19 a) With the help of a suitable example, show the working of Diffie-Hellman key exchange algorithm. (9)
b) Explain about Secure Socket Layer (5)

OR

- 20 a) Describe the significance of Orthogonal Frequency Division Multiplexing (OFDM) in next generation networks (4)
b) Discuss about various security issues in mobile computing (10)
